

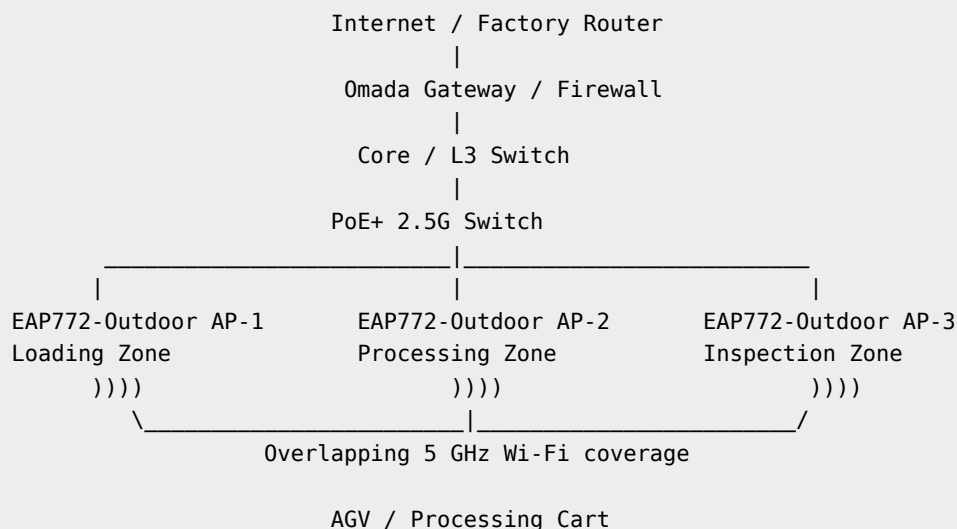
TP-Link Omada EAP772 AP + Dongle client

1. Assumed smart-factory simulation

Factory area: Processing line / production lane
Route length: 70–90 m
Route width: 10–15 m
Mobile client: AGV or processing cart with industrial PC + Wi-Fi dongle
AP: TP-Link Omada EAP772-Outdoor
Controller: Omada Software Controller / OC200 / OC300
Main traffic: MES commands, AGV status, barcode/scanner data, process events, light diagnostics
Not recommended for: safety-critical emergency stop or hard real-time PLC control

The EAP772-Outdoor is suitable for this budget simulation because it is a tri-band Wi-Fi 7 outdoor AP with **1×2.5G PoE+ port**, **IP68 enclosure**, **Omada centralized management**, **mesh**, and **seamless roaming** support. TP-Link lists recommended coverage at around **300 m² / 3,200 ft²**, but in a metal-heavy factory you should design more conservatively with overlapping APs. ([Omada](#))

2. Simulated topology



Industrial PC + Wi-Fi Dongle
|
MES / PLC Gateway / ERP

For your first pilot, I would use **3 APs for one processing lane**. If the lane is long, blocked by metal machines, or has multiple turns, use **4 APs**.

3. AP placement simulation

Processing route

Material Input → Processing Machine → Inspection → Output / Packaging
AP-1 AP-2 AP-3 AP-4 optional

AP	Location	Purpose
AP-1	Material input / loading area	First connection point for AGV/cart
AP-2	Main processing machine area	Main process communication zone
AP-3	Inspection / QC area	Roaming target from AP-2
AP-4 optional	Packaging/output area	Needed if output area is far or blocked by metal

Practical spacing

Item	Recommended value
AP spacing in factory	20–30 m starting point
Coverage overlap	25–35%
Target RSSI on route	-55 to -67 dBm
Minimum acceptable RSSI	Avoid going below -70 dBm
Preferred band	5 GHz
Channel width	20 MHz first , 40 MHz only if clean
6 GHz	Use only if dongle supports it and local regulation allows it

Even though the EAP772-Outdoor supports 2.4 GHz, 5 GHz, and 6 GHz, I would build the AGV/process SSID mainly on **5 GHz** because most industrial dongles support 5 GHz better than 6 GHz, and 2.4 GHz is usually noisy in factories.

4. VLAN and IP design

Use separate networks for management, AGV/process traffic, PLC/machine traffic, and cameras.

Network	VLAN	Example subnet	Purpose
AP management	VLAN 10	10.10.10.0/24	Omada APs, controller
AGV / processing Wi-Fi	VLAN 20	10.20.20.0/24	Dongle clients
PLC / machine network	VLAN 30	10.30.30.0/24	PLC gateway, machine controllers
Camera / diagnostics	VLAN 40	10.40.40.0/24	Optional video/sensor traffic
Guest / maintenance	VLAN 90	10.90.90.0/24	Maintenance laptops only

Example IP plan:

```
Omada Controller:    10.10.10.10
AP-1:                10.10.10.21
AP-2:                10.10.10.22
AP-3:                10.10.10.23

AGV-01 Wi-Fi dongle: 10.20.20.101
AGV-02 Wi-Fi dongle: 10.20.20.102
MES server:          10.20.20.10
PLC gateway:         10.30.30.10
```

5. SSID setup

Create one dedicated production SSID for the dongle clients.

Setting	Recommended value
SSID	FACTORY_PROCESS_AGV

Setting	Recommended value
VLAN	20
Band	5 GHz only for first simulation
Security	WPA2/WPA3 Enterprise if possible
Pilot security	WPA2-PSK acceptable only for testing
Roaming	Enable Omada fast roaming only after dongle compatibility test
Band steering	Disable if using 5 GHz-only SSID
Load balancing	Disable for AGV/process clients
Minimum RSSI	Use carefully; do not force disconnect too aggressively
QoS	Prioritize control/status packets over diagnostics/video

TP-Link explains that Omada fast roaming uses **802.11k/v/r concepts**, but these features only work when the **client device also supports them**. This is important because the Wi-Fi dongle, not the AP, usually makes the final roaming decision. ([TP-Link](#))

6. Channel plan

Use fixed channels first instead of full auto during the simulation.

AP	5 GHz channel	Channel width	Tx power
AP-1	36	20 MHz	Medium
AP-2	44	20 MHz	Medium
AP-3	149	20 MHz	Medium
AP-4 optional	157	20 MHz	Medium

Avoid setting all APs to maximum power. Too much AP power can create a **sticky-client problem**, where the dongle remains connected to a far AP instead of roaming to the closer AP.

7. Dongle-side configuration

The dongle is the weak point, so configure it carefully.

Dongle setting	Recommendation
Wi-Fi standard	Wi-Fi 6 or Wi-Fi 7 preferred
Band	5 GHz preferred
Antenna	Use external antenna if AGV body is metal
Power saving	Disable
USB selective suspend	Disable
Roaming aggressiveness	Medium or medium-high, then test
Driver updates	Freeze driver version after validation
IP addressing	DHCP reservation preferred
Watchdog	Add reconnect script/service

For Windows industrial PC:

```
Device Manager
→ Wi-Fi Adapter
→ Power Management
→ Uncheck "Allow the computer to turn off this device to save power"
```

For Linux industrial PC:

```
sudo iw dev wlan0 set power_save off
```

8. Processing workflow simulation

1. AGV enters material input area.
2. Wi-Fi dongle connects to AP-1 through SSID FACTORY_PROCESS_AGV.
3. MES sends job order to AGV.
4. AGV moves toward processing machine.
5. AP-1 signal decreases; AP-2 signal increases.
6. Dongle roams from AP-1 to AP-2.
7. AGV reaches processing station.
8. Industrial PC sends "arrived at station" event to MES.
9. PLC gateway confirms machine-ready status.
10. Processing starts.
11. AGV/cart sends status every 500 ms to 1 second.
12. After processing, AGV moves to inspection zone.
13. Dongle roams from AP-2 to AP-3.

14. Inspection result is sent to MES.
15. AGV moves to output/packaging zone.

Expected roaming signal pattern:

At loading area:

AP-1: -55 dBm

AP-2: -72 dBm

Between loading and processing:

AP-1: -66 dBm

AP-2: -62 dBm

At processing station:

AP-1: -75 dBm

AP-2: -55 dBm

AP-3: -68 dBm

The target is that the AGV always sees **at least two usable APs** before roaming is needed.

9. Hardware list for one pilot lane

Item	Quantity	Notes
TP-Link Omada EAP772-Outdoor	3-4	APs for overlapping coverage
Omada OC200 / OC300 or software controller	1	Required for centralized management and roaming features
2.5G PoE+ switch	1	EAP772-Outdoor has 1×2.5G PoE+ RJ45 and supports 802.3at PoE. (Omada)
Industrial PC / mini PC	1-3	Installed on AGV/cart
Wi-Fi 6/7 dongle	1 per client	Prefer external antenna
Shielded Cat6A cable	Per AP	Factory EMI protection
UPS	1	For PoE switch/controller
MES / process server	1	Simulation server

10. Validation test plan

Run these tests before using it in a real production process.

Test	Expected result
Walk/drive AGV from AP-1 to AP-2 to AP-3	No application disconnect
Continuous ping to MES	No long packet loss during roaming
Reboot AP-2 during route	Client reconnects to AP-1 or AP-3
Start nearby motors/machines	Retry rate should not spike heavily
24-hour soak test	No dongle sleep/disconnect
3–5 clients moving together	Stable latency and acceptable channel utilization
Process event test	"Arrived," "Started," "Completed" events delivered reliably

Useful test command:

```
ping 10.20.20.10 -i 0.2
```

Application-level test:

```
Send AGV/process status every 500 ms.  
Record latency, packet loss, reconnect count, roaming count, and missed process events.
```

11. Final recommended simulation setup

```
AP model: TP-Link Omada EAP772-Outdoor  
Client: Industrial PC + Wi-Fi 6/7 dongle  
AP count: 3 for short processing lane, 4 for longer/blocked lane  
Controller: Omada OC200/OC300 or Omada Software Controller  
SSID: FACTORY_PROCESS_AGV  
VLAN: 20  
Band: 5 GHz  
Channel width: 20 MHz  
AP overlap: 25–35%  
Target RSSI: -55 to -67 dBm  
Cable: Shielded Cat6A
```

Switch: 2.5G PoE+ switch with UPS
Security: WPA2/WPA3 Enterprise preferred

This is a strong **low-cost smart-factory Wi-Fi pilot**. The main engineering risk is not the EAP772-Outdoor AP; it is the **dongle chipset, antenna placement, roaming behavior, and driver stability**.